Q.1 How are frequency and period related in simple harmonic motion?

A. They are inversely related

B. Their sum is constant

C. They are directly related

D. None of the above

Q.2 Consider a mathematical statement, V = IR, where V be the constant of proportionality and it represents the voltage. The equation is called _____.

A. Coulomb's law

B. Ohm's law

C. Hooke's law

D. Newtown's law

Q.3 The damping force is _____to the instantaneous velocity \frac{{dx}}{{dt}}

A. None of these

B. Inverse proportional

C. Constant

D. Proportional

Q.4 Solution of the D.Equation 4y^{//} + y = 0. is

A. None of them.

B. y(x) = c_1 {x} {2}

C. y(x) = c_1 Sin{x} {2}

D. y(x) = c_1 {x} {2} + c_2 Sin{x} {2}

Q.5 Vibration of an object on equilibrium point is called simple harmonic motion when the restoring force is proportional to _____.

A. A spring constant

B. Time

C. Mass

D. Displacement

Q.6 {\text{The}}\,\,{\text{quantity}}\,\,Z = \sqrt {{X^2} + {R^2}} \,\,{\text{is}}\,\,{\text{called}}\,\,__________\,\,{\text{of}}\,\,{\text{the}}\,\,{\text{circuit}}{\text{.}}

A. {{impedance}}

B. {{reactance}}

Q.7 \begin{gathered} {\text{In}}\,\,{\text{the}}\,\,{\text{study}}\,\,{\text{of}}\,\,{\text{mechanics,}}\,\,{\text{cosider}}\,\,{\text{the}}\,\, - \beta \,{\left({\frac{{dx}}{{dt}}} \right)^2},\,{\text{where}}\,\,\beta \,\,{\text{is}}\,\,{\text{a}}\,\, \hfill \\ {\text{damping}}\,\,{\text{constant}}\,\,{\text{and}}\,\,{\text{negative}}\,\,{\text{sign}}\,\,{\text{indicates}}\,\,{\t \hfill \\ {\text{to}}\,\,{\text{the}}\,\,{\text{direction}}\,\,{\text{of}}\,{\text{motion}}{\text{.}}\,\,\,\, \hfill \\\ \end{gathered}

A. {{opposite}}

B. {{same}}

Q.8 {\text{The}}\,\,{\text{power}}\,\,{\text{series}},\,\,1 + x + \frac{x}{{2!}} + \frac{{{x^3}}}{{3!}} + ...\, = _____.

A. {e^x}

B. x

C. (1 + x)

D. x

Q.9 The periodic time is given by

A. 2 *

B. { }{ }

C. { }{{2 }}

D. {{2 }}{ }

Q.10 The hook's law states that the force F is proportional to the _____.

A. Elongation

B. Length

C. Weight

D. None of these
